

Week 4 HW Question 4:

Date:

$$A = \begin{bmatrix} 4 & 3 & 0 & 0 & 4 \\ 0 & 1 & 2 & 2 & 3 \\ 0 & 0 & 3 & 6 & 7 \\ 0 & 0 & 4 & 5 & 5 \\ 0 & 0 & 0 & 0 & 7 \end{bmatrix}$$

$$\det(A) = 7(-1)^{5+5} \begin{vmatrix} 4 & 3 & 0 & 0 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 3 & 6 \\ 0 & 0 & 4 & 5 \end{vmatrix}$$

$$= 7 \left[4(-1)^{1+1} \begin{vmatrix} 1 & 2 & 2 \\ 0 & 3 & 6 \\ 0 & 4 & 5 \end{vmatrix} \right]$$

$$= 28 \left[1(-1)^{1+1} \begin{vmatrix} 3 & 6 \\ 4 & 5 \end{vmatrix} \right]$$

$$= 28 \begin{vmatrix} 3 & 6 \\ 4 & 5 \end{vmatrix}$$

$$= 28[(3 \cdot 5) - (4 \cdot 6)]$$

$$= 28[15 - 24] = 28(-9) = -252$$